Review of the pmsimstats-ng Whitepapers: Notation, Accuracy and Conventions

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Purpose. This document records a deep review of the 34 numbered whitepapers in `docs/` (09a, 19 to 51, and `whitepaper-covar-vs-orig-dgp.md`, renumbered `docs/54` on 2026-10-09) for three things: consistency with `analysis/report/NOTATION.md`, accuracy, and the project's documentation conventions. Problems found were fixed in place; superseded statements received dated notes rather than silent rewrites. It lists what changed in each document and the items left open.

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Notation index and glossary

Beyond symbol names quoted from the reviewed documents, this document uses only the following symbols of `analysis/report/NOTATION.md` (index added 2026-10-09).

Table 1: Notation index of symbols used in this document

Symbol	Meaning	Status
η_t, η	design expectancy at visit t (1 open label, 0.5 blinded, 0 at baseline); η as the 2026-10-07 `NOTATION.md` wrote it (Section 2)	canonical
e_t	former symbol for η_t, quoted only to record the rename	retired
ρ	AR(1) serial correlation per week	canonical
ρ_{cs}	common correlation of the compound-symmetry construct (row added 2026-10-09)	canonical
N	total sample size across randomization paths	canonical

Glossary.

- **Dated note.** A short paragraph, dated 2026-10-07, that records a correction or a superseding result and points to the newer document.
- **Linter.** `tools/notation-lint.pl`, which checks a few mechanical rules of `NOTATION.md` (bare `Dbc` and `Sx`, legacy specification labels, day-denominated half-lives, per-path values of N).

1. Method

Each document was read end to end against one rubric:

- **Notation.** Every symbol checked against `NOTATION.md`; colliding or non-canonical symbols renamed (not merely declared as departures); a notation index and glossary placed at the front; the linter run before and after.
- **Accuracy.** Derivations re-derived; numbers in prose, tables and captions cross-checked; claims about code checked against the current source and, for the published process, commit `58b32a9`; numerical claims checked against the cited data files and scripts; references checked against PubMed or Crossref.

- **Conventions.** Header block (title, timestamp, Author, Purpose), contents block and captions, US English, no em dashes, evidence labels, no hand-written render stamps, single titles, figure captions not duplicated, resolvable citations.

Every document was rendered through `tools/render.sh` from `docs/` and checked with `pdftotext` and `pdfimages`.

Final state (verified). The linter is clean on all 34 documents. No em dashes, no hand-written "Rendered on" lines and no use of the word "epistemic" remain. Every table carries a caption, and every figure is embedded and listed (docs/50, the two-page brief, has no contents block by design).

2. Changes that apply across documents

- **Expectancy symbol.** `NOTATION.md` defines η as the canonical expectancy covariate. The design expectancy written e_t (1 open label, 0.5 blinded, 0 at baseline; code `e`, regressor `De`) is now η_t in every document. In `docs/46` the rename is recorded in Amendment 2 as typographical. (Notation, 2026-10-09: `NOTATION.md` now defines η_t itself as the design expectancy and reserves $\tilde{\eta}_{it}$ for a measured placebo-belief covariate.)
- **Ceiling values.** Statements no longer mix serial correlations: the compound-symmetry ceiling is 0.256 without carryover; the separable AR(1) ceiling is 0.440 at $\rho = 0.8$ and 0.474 (Hybrid 0.480) at $\rho = 0.7$. The published process uses compound symmetry with $\rho_{cs} = 0.8$ (`docs/34`). (Notation, 2026-10-09: the compound-symmetry correlation was written ρ.)
- **Stress-test outcome.** Documents that describe the stress test now give 41 cells, 28,000 trials and the 14-analysis decision rerun (`docs/46`, Amendment 2): F1 selected, the phase-variance versions G1 and G2 effectively equivalent, F1A failing size.
- **Superseded recommendations.** The unstructured Kenward-Roger recommendation of `docs/45` and its repetitions are marked superseded by `docs/46` and `docs/49`. The flexible-time size excess is stated as real under model-based and Kenward-Roger inference and absent under CR2 (`docs/42`).
- **References.** Every reference list was checked. Several documents had copied invented author lists from `docs/references/nof1-simulation.bib` (corrected separately the same day); these are now attributed to the real papers (for example Shi and Ye 2024, Chen and Chen 2014, Hawksworth et al. 2024, Chatters et al. 2024, Senn 2019, Lim and In 2021, Schmid and Yang 2022). DOIs were added where they exist.

3. Per-document summary

Counts are approximate; each count is a group of related edits.

Table 2: Review outcome by document

Document	Notation	Accuracy	Main accuracy findings
09a	6	13	Headline "drop then recovery" not supported (within 2 MCSE); model misdescribed; three of six references wrong; conclusion 4 withdrawn
19	6	2	Noise SD 5 corrected to 8; shift 650 corrected to 558
21	10	7	Hybrid loss 68% corrected to 39%; range "40-60%" corrected to 20-61%; tables not reproducible
22 (journal)	3	9	Six misattributed citations corrected
22 (parity)	1	6	Duplicate title removed; script behavior corrected against the code
23	25	17	Multiplicative/additive equivalence, AR(1) residual formula, step-rule row label, seven references
24	8	13	Gompertz parameter meaning reversed; Hybrid design misdescribed; variance identity
25	8	10	Parameter table (TV row) wrong; heterogeneity claim withdrawn; superseded by paper 07
26	12	9	Sphericity conflated with compound symmetry; misattributed Type I figure
27	8	9	Tier 1 cell count 264 corrected to 324; E2 decision premise false; duplicate title
28	10	3	Superseded by docs/31 (light review); crossover row and power pair corrected
29	0	2	US English; two definitions corrected
30	3	4	Stale process statements updated
31	15	22	Gompertz section; covariance dimension; paper 06 sample sizes; useDE claim; references
32	4	8	Commit dating; "one-week" collapse corrected to any positive half-life; ceiling decline with visits
33	6	3	One value 20.0% corrected to 19.9%; CS versus AR(1) reversal attributed to step coupling

Document	Notation	Accuracy	Main accuracy findings
34	2	8	Published-process numbers used $\rho = 0.7$, not 0.8; binding path A, not C; second silent repair
35	9	11	Ceiling formula term; stress-test description (26,500 to 28,000; not nominal everywhere)
36	8	7	Five configurations and 261,000 fits corrected to six and 309,000
37	4	7	Six small numerical misquotes corrected against the data files; derivations verified
38	12	4	Sign of a correlation corrected; Hobbs 1978 subjects
39	8	8	48 conditions corrected to 96; noise ratio a fifth, not a quarter
40	1	5	Figure references swapped; time on drug restarts at zero
41	6	6	Power floor range; recommendation superseded
42	3	8	MCSE multiples; flexible-time correction applied
43	2	20	14-analysis stress-test table added; status entries updated
44	3	9	"Carryover raises every ceiling" false at $\rho = 0.5$; step statuses dated
45	6	7	Section 4.4 correction; superseded primary recommendation
46	5	4	Amendment 2 added; original protocol text unchanged in substance
48	3	7	Pending results filled in; maximin table scope stated
49	9	8	Pending results filled in; cancellation claim qualified
50	1	5	Analysis table and recommendation updated; still two pages
51	18	7	Package carryover is anchored, not compounding; Feature 6 results; dropout size wording
54 (covar vs orig)	100	13	"orig" means the 2024 copy, not `58b32a9`; AR(1)-alone result depends on an uncoupled first visit

4. Open items

Code.

- `R/carryover_analysis.R`: `threshold_bm` and `predicted_effect` added `bm_mean`, but `lme_analysis()` fits the uncentered biomarker, so the threshold was offset. Fixed 2026-10-08: both are now computed on the fitted scale, and `inst/tinytest/test_carryover_analysis.R` checks the predicted effect against the model's own contrast (verified; with a biomarker of mean 98 the old code was off by 7.0 points and put the threshold at 172.6 instead of 74.5).
- `R/generateData.R:391-393`: the Cholesky fallback still repairs a non-positive-definite matrix silently.
- `analysis/scripts/carryover-sensitivity/simulation-core.R`: a comment says the TV parameters follow the published defaults; they do not.

Data and reproducibility.

- `docs/21`: the tables cannot be reproduced; the driver's output file is missing. Re-run the driver at a realistic replicate count, or retire the tables in favor of paper 01.
- `analysis/scripts/parity/`: the committed `parity_report.rds` is a quick-mode report (16 rows) while `parity_baseline.rds` is default mode (144 rows), so `parity-diff.R` compares unlike reports.
- `docs/34`: $\rho = 0.8$ is missing from `rho.csv` and Figure 2; the added table row was computed independently. Figure 10's subtitle ("binding paths") is wrong for Hybrid and needs the figure script.
- `docs/40`: the figure files `40-fig2` and `40-fig3` are numbered opposite to the order in which they print.
- Manuscript 02: whether its crossover cells were rerun with the corrected exposure coding is not established.

Documents.

- `docs/47`, the stress-test results report referred to by `docs/46` and `docs/48`, did not exist; it was written on 2026-10-08 from the stored decision outputs.
- The 14-analysis rerun ranks G1 (0.308) and G2 (0.312) ahead of F1 (0.358) over all 26 alternative cells, prognostic cells included. The working protocol does not say whether those cells count; F1 remains selected.
- `docs/09a`: the source of its Section 1.1 table is unknown, and its robustness recommendation cites simulations it does not report.

- `docs/24`, `docs/31`: three claims have no source and are flagged in the text (residual expectancy of 30-60%, a chronic-fatigue example, convergence below 90%). Corrections made in `docs/31` to Parts I-III should be carried into `docs/24`.
- `docs/28`, `docs/31` and paper 06 may still quote stale per-path sample sizes for Studies A and C.
- `docs/20-annotated-bibliography.tex` probably repeats the reference errors corrected in `docs/23`; not checked.
- The AHRQ user-guide chapter authors rest on web-search reports; the AHRQ site blocks automated access.

Not done. No Monte Carlo cell was re-simulated; numerical checks used stored outputs. The `.tex` companions of the documents were not reviewed.

5. Reproducibility

The per-document review logs and check scripts were written to the session scratchpad and are not part of the repository. The final checks can be repeated with:

```
perl tools/notation-lint.pl docs/[0-9]*.md
```